Pharma Product Analytics & Market Intelligence

Problem Statement

Pharmaceutical companies and healthcare analysts need to **optimize pricing, understand drug composition trends, and identify market opportunities** across therapeutic categories.

Using the given pharma dataset, analyze **product-level data** to uncover insights on **drug composition, pricing patterns, manufacturer dominance, and therapeutic segmentation** to support **data-driven pharma strategy decisions**.

Project Objectives

- Analyse pharma product portfolio across manufacturers
- Identify pricing trends based on drug composition and dosage
- Understand therapeutic class distribution
- Evaluate multi-ingredient vs single-ingredient drugs
- Detect market patterns for strategic decisions

Key Tasks

1. Data Understanding & Preparation

- **Explore dataset structure and fields**
- **Handle missing or inconsistent values**
- **Standardize fields like dosage form, strength, and packaging**
- **Parse and extract information from active_ingredients**

2. Product Portfolio Analysis

- **Count total products and unique manufacturers**
- **Analyze distribution of dosage forms (tablet, capsule, etc.)**
- **Analyze pack size and packaging types**
- **Identify active vs discontinued products**

3. Therapeutic Class Analysis

- **Count of products by therapeutic class**
- **Identify top therapeutic categories (e.g., antibiotics)**
- **Analyze product diversity within each class**

4. Drug Composition Analysis

- **Classify single vs multi-ingredient drugs**
- **Identify most common primary ingredients**
- **Analyze frequently occurring drug combinations**
- **Compare strength variations (e.g., 250mg vs 500mg)**

5. Pricing Analysis

- **Analyze overall price distribution**
- **Compare price across therapeutic classes**
- **Analyze price vs:**
    - **Pack size**
    - **Number of active ingredients**
    - **Dosage form**
- **Identify pricing outliers**
- **Compare price variation for same ingredient/strength**

6. Manufacturer Analysis

- **Identify top manufacturers by product count**
- **Compare pricing across manufacturers**
- **Analyze manufacturer focus by therapeutic class**

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7. Product Segmentation

- **Segment products into price categories (low / mid / premium)**
- **Segment based on number of ingredients**
- **Segment based on dosage form**

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8. Descriptive Insight Generation

- **Identify most common:**
    - **Therapeutic class**
    - **Ingredient**
    - **Dosage form**
- **Analyze distribution patterns across:**
    - **Price**

- **Pack size**
- **Manufacturers**

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9. Insight-Based Questions to Answer

• Which therapeutic class has the highest number of products?
• Which dosage form dominates the dataset?
• Do combination drugs tend to be priced higher?
• Which ingredients are most frequently used?
• How does price vary with pack size and strength?
• Which manufacturers have the largest product portfolio?
• Are there pricing inconsistencies for the same drug composition?

Deliverables

1. Analytical Report

• Problem understanding
• Key findings from each analysis section
• Business recommendations

2. Dashboard (Power BI / Tableau)

Include:
• Product distribution by therapeutic class
• Pricing analysis visuals
• Manufacturer comparison
• Drug composition insights

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3. Insight Summary (Presentation Ready)

• 8–10 key business insights
• Strategic recommendations such as:

- Pricing optimization
- Product portfolio gaps
- High-demand drug categories